

Quality Comparison of Compressed and Uncompressed Music

Final Report

Author: Oliver Fischer

I. Abstract

Digital music is stored in either a compressed or uncompressed audio format. This project analyzed both audible and measurable properties of WAV files (uncompressed) and compared them to FLAC, M4A, MP3 320, 256, 192, 128 and 64 (compressed). Three songs of distinct genres were used: You Can Never Be Ready – Sunrise Avenue (Rock), Emoji – Pegboard Nerds (Electronic) and a nameless royalty-free orchestral piece (Classical). Each file underwent a spectral analysis in Audacity with the results being recorded. The data of compressed music was compared to the data of uncompressed music and differences were notated.

Only one reliable measurable difference could be observed, which is the removal of higher frequency content in MP3 and M4A files. Higher compression rates lead to more higher frequencies being cut off to reduce the amount of information stored, leading to smaller file sizes. However, subjective hearing tests revealed that other frequencies, especially mids, are also altered during the complicated MP3 compression process, resulting in less detailed music. The most efficient way of storing music is MP3 320, which drastically reduces file size while remaining essentially indistinguishable from FLAC.

II. Introduction and Background

At the heart of this project lies the concept of file compression, which has the simple goal of making file sizes smaller by reducing the amount of bits compared to the original file. To achieve this, there are generally two paths: lossy compression and lossless compression¹. File compression is a general concept. This project focused only on audio data compression. Lossless compression preserves the quality of audio files by only removing things like statistical redundancies (for example, periods of silence). Lossy compression on the other hand uses specific compression algorithms (audio codecs) to remove certain data and thus reduce the file size dramatically. Ideally, when it comes to music file compression, the algorithms would remove information in such a way that humans would not or barely be able to perceive any differences in audio quality². Arguably, some formats like 320kbit/s MP3's have achieved that.

The algorithms are based on psychoacoustics, which is the study of the perception of sound. Information that is deemed less important or less audible is usually targeted for compression³. Due to the internet, MP3 files have become the widely used standard thanks to their small file sizes⁴. M4A uses the AAC audio codec, which offers better sonic quality and was designed to succeed MP3, but was never able to do so. FLAC offers lossless compression and should, in theory, not have any audible

¹WhatIs.com., (2016). *lossless and lossy compression*. <http://whatis.techtarget.com/definition/lossless-and-lossy-compression> Accessed 5th October 2016.

²Oral History in the Digital Age., (2016). *Compression*. <http://ohda.matrix.msu.edu/gettingstarted/glossary/videoglossary/compression/> Accessed 5th October 2016.

³Ibid.

⁴Universal Audio., (2016). *Understanding Audio Data Compression: MP3s, AACs, and more*. <http://www.uaudio.com/blog/understanding-audio-data-compression/> Accessed 5th October 2016.

difference compared to uncompressed audio files, but at the expense of larger file sizes than MP3 and M4A⁵.

III. Goals of the Experiment

The goal of this experiment was to discover concrete measurable differences between compressed and uncompressed music as well as look at how these translate into the perceived sound of music. In addition, this project also aimed to showcase the significance of file compression in digitally stored music and increase awareness of the benefits of higher quality music.

IV. Procedure and Equipment

Three songs were used for this project. The WAV file for ‘You Can Never Be Ready’ was acquired from a CD album. ‘Emoji’ was downloaded as a WAV file from *Bandcamp* and the orchestral piece was downloaded as a WAV file from *dl-sounds.com*.

All three files were imported in Audacity and exported as FLAC, M4A, MP3 320kbit/s, 256kbit/s, 192kbit/s, 128kbit/s and 64kbit/s. All MP3 files have a constant (rather than variable) bit rate to ensure that they are comparable. Furthermore, they were all exported in joint stereo mode.

All of the files, from WAV to MP3 64, were then individually imported in Audacity and the built-in frequency analysis tool was used to analyze the length of the entire song. To provide the most detail, the size of the analysis was set to 65536. A screenshot of each analysis was then made.

⁵Ibid.

The frequency diagrams of the WAV encoded songs were used as a reference. The diagrams of the compressed songs in each respective genre were compared against the references in that genre and differences were notated.

A subjective hearing test was also conducted using a Focusrite Scarlett 2i4 audio interface that was connected to isolated JBL LSR305 monitors via balanced TRS cables and with all room controls set to 0dB. Beyerdynamic DT880 Pro studio headphones connected to the same interface were used to monitor bass response. Comparisons between different MP3 compression rates and FLAC were conducted. FLAC was compared to WAV. 256Kbit MP3 was then compared to M4A. FLAC was also compared to MP3 320 and 256. The sonic quality of highly compressed music (MP3 192 to 64) was also observed and put in words.

In an attempt to measure further differences, a low-pass filter was applied to two of the WAV songs, cutting off all frequencies above 16.5KHz, which is roughly equivalent (slightly worse) than a 128kbit/a MP3 file. A frequency analysis and hearing tests were then conducted on the resulting files.

V. Summary of Data Collected

a. Raw Data

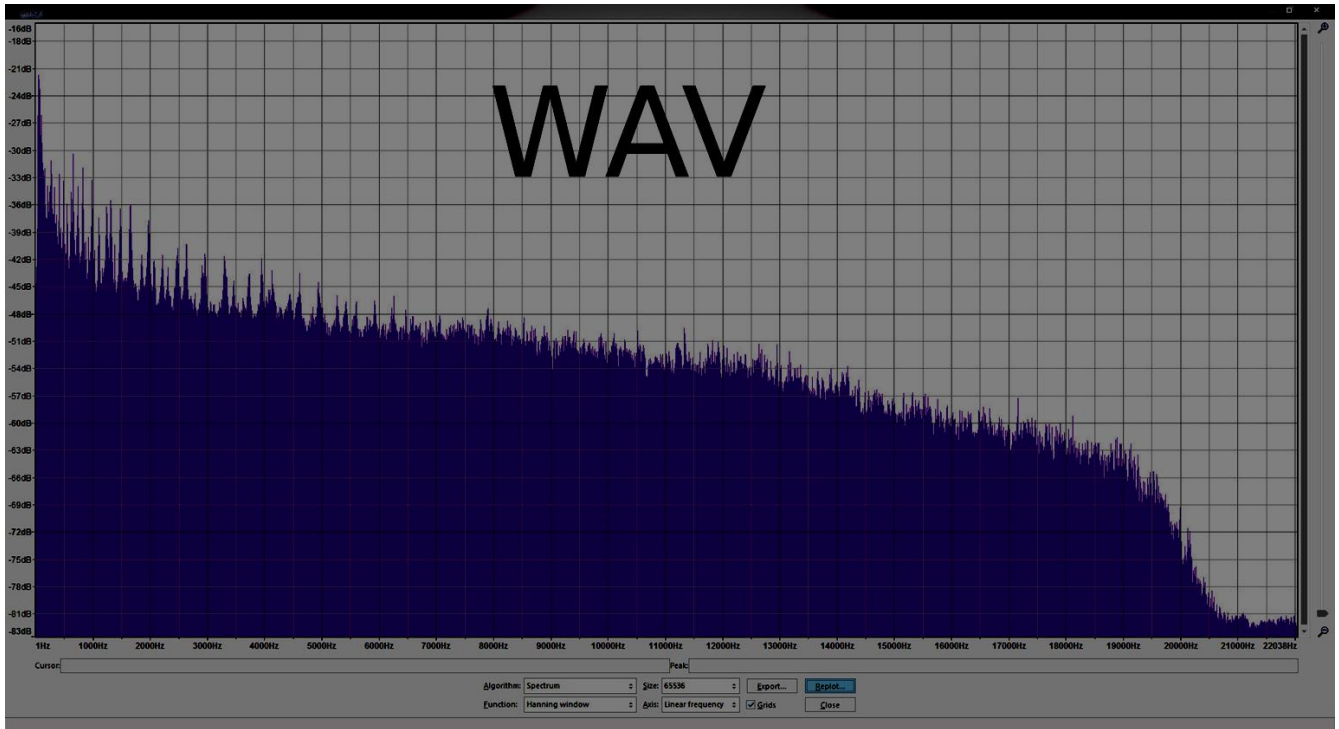
In the following table, WAV is always used as a reference.

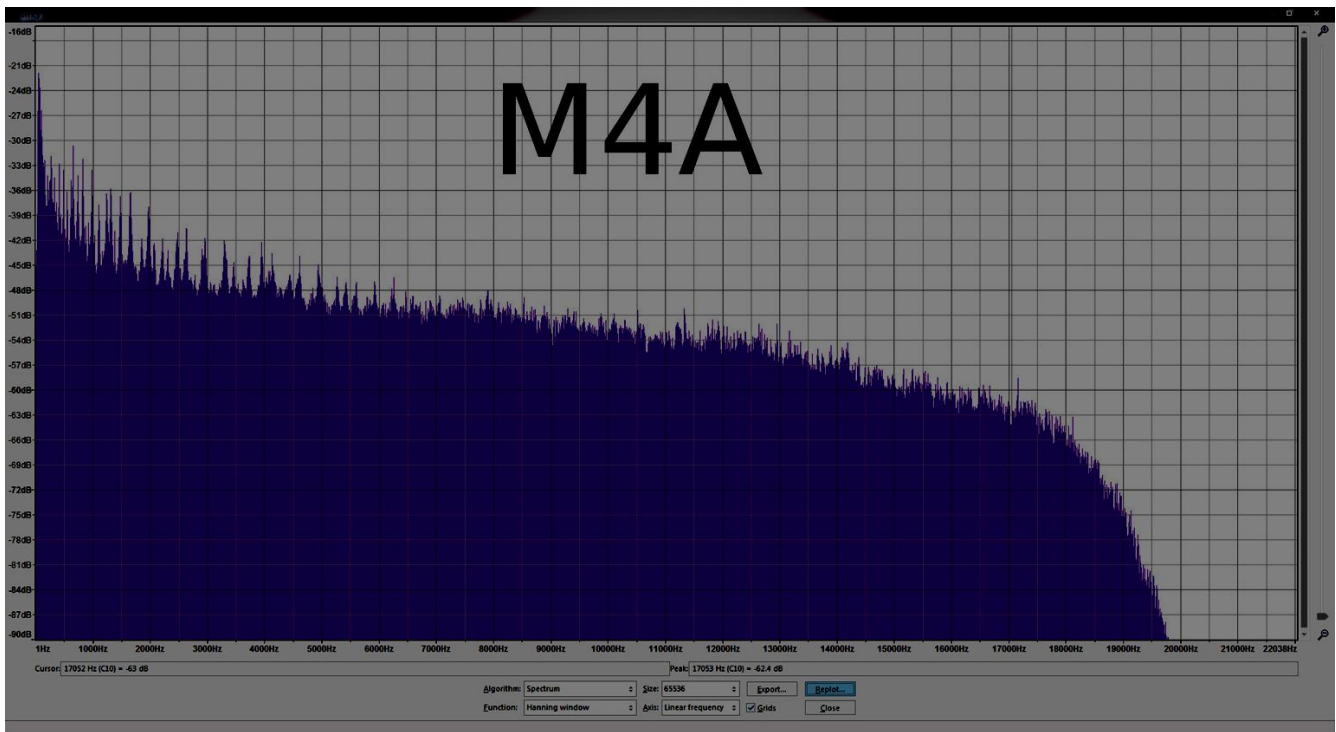
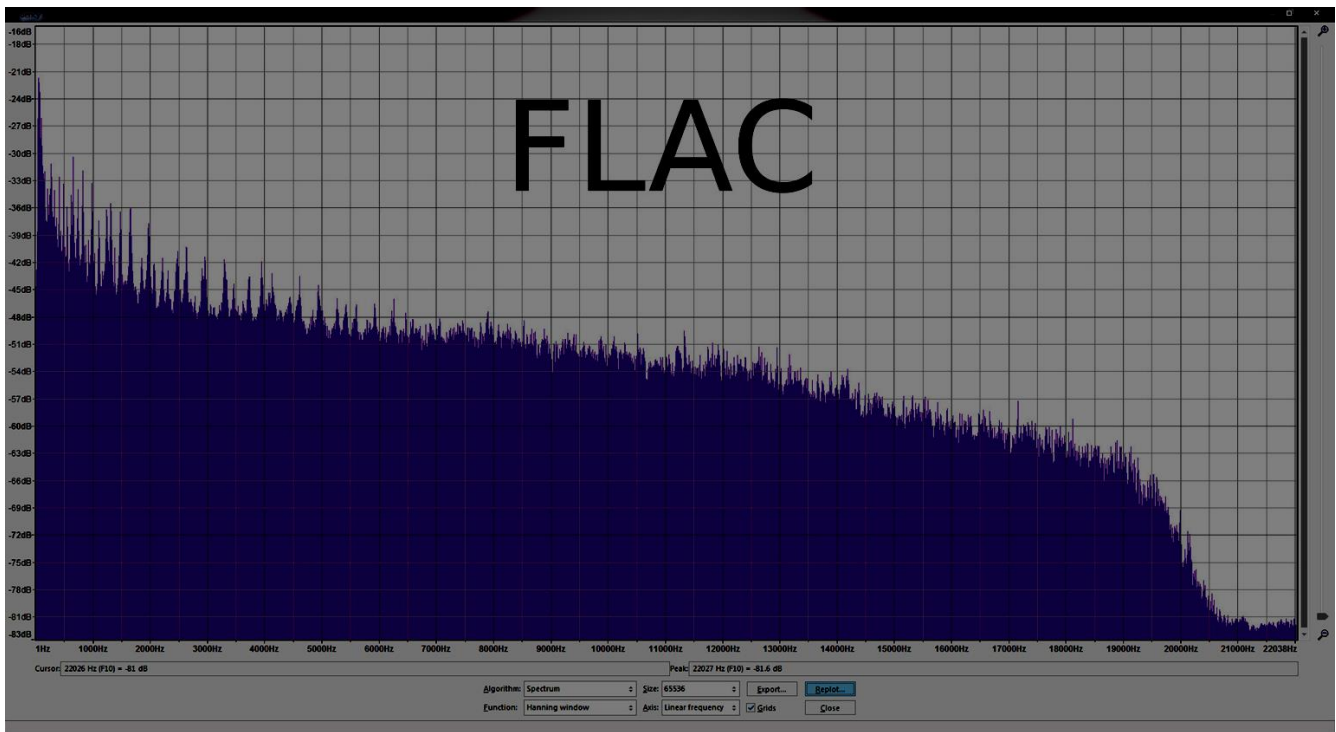
File type	Cut-off frequency (measured)	Perceivable difference (in theory)⁶	Perceivable difference (subjective hearing test)	File size ‘Emoji’	Extrapolation for 2500 song library
WAV	N/A	N/A	N/A	38.8 MB	97 GB
FLAC	None	No	No	30.9 MB	77.25 GB
M4A	19750Hz	No	Yes	13.4 MB	33.5 GB
MP3 320	20100Hz	No	No ⁷	8.8 MB	22 GB
MP3 256	19500Hz	No	Yes	7 MB	17.5 GB
MP3 192	18750Hz	Yes	Yes	5.3 MB	13.25 GB
MP3 128	16700Hz	Yes	Yes	3.5 MB	8.75 GB
MP3 64	11200Hz	Yes	Yes	1.8 MB	4.5 GB

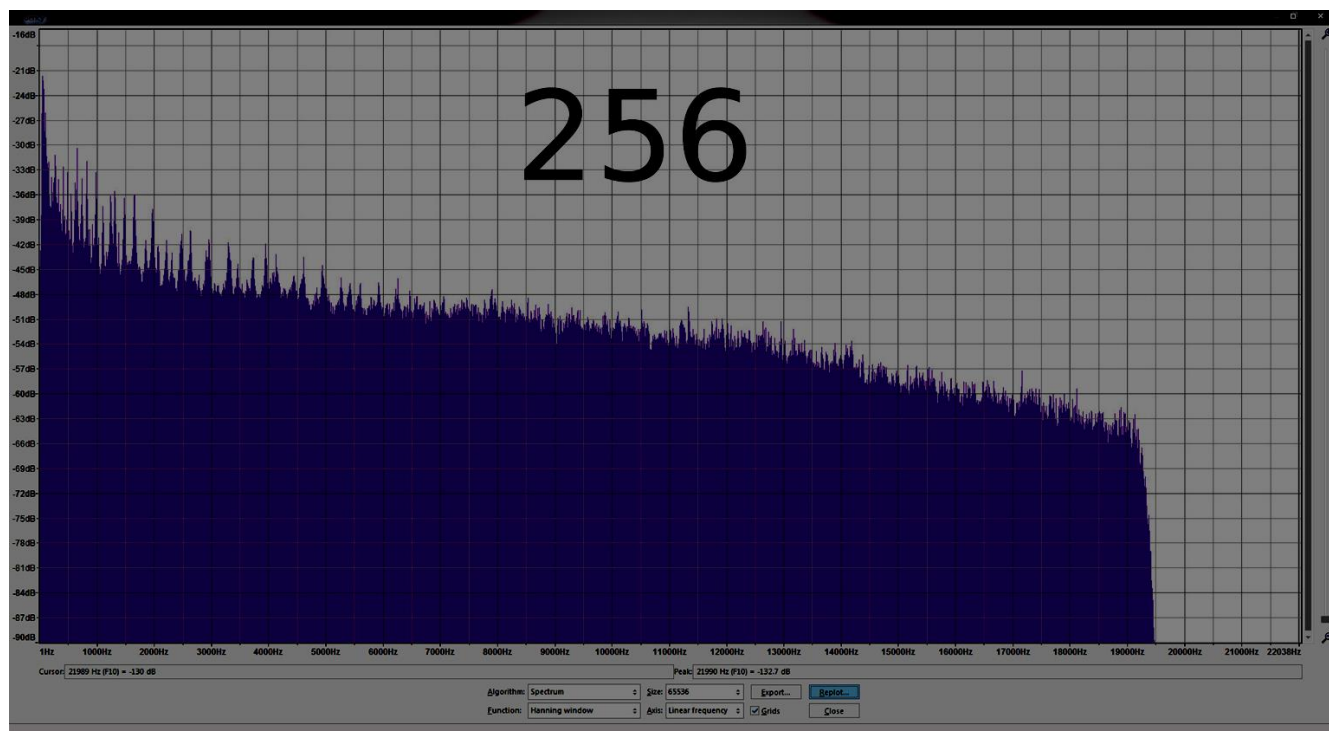
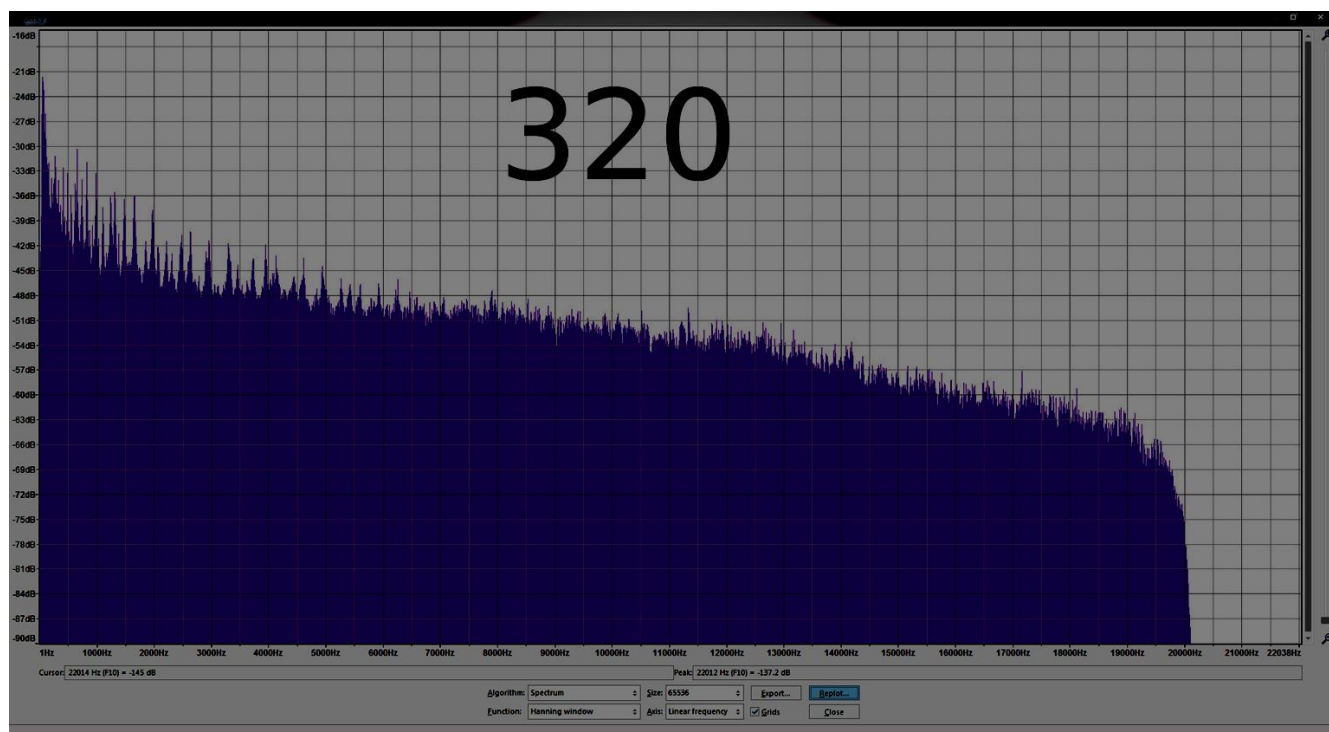
b. Full example of ‘Emoji’ Diagrams

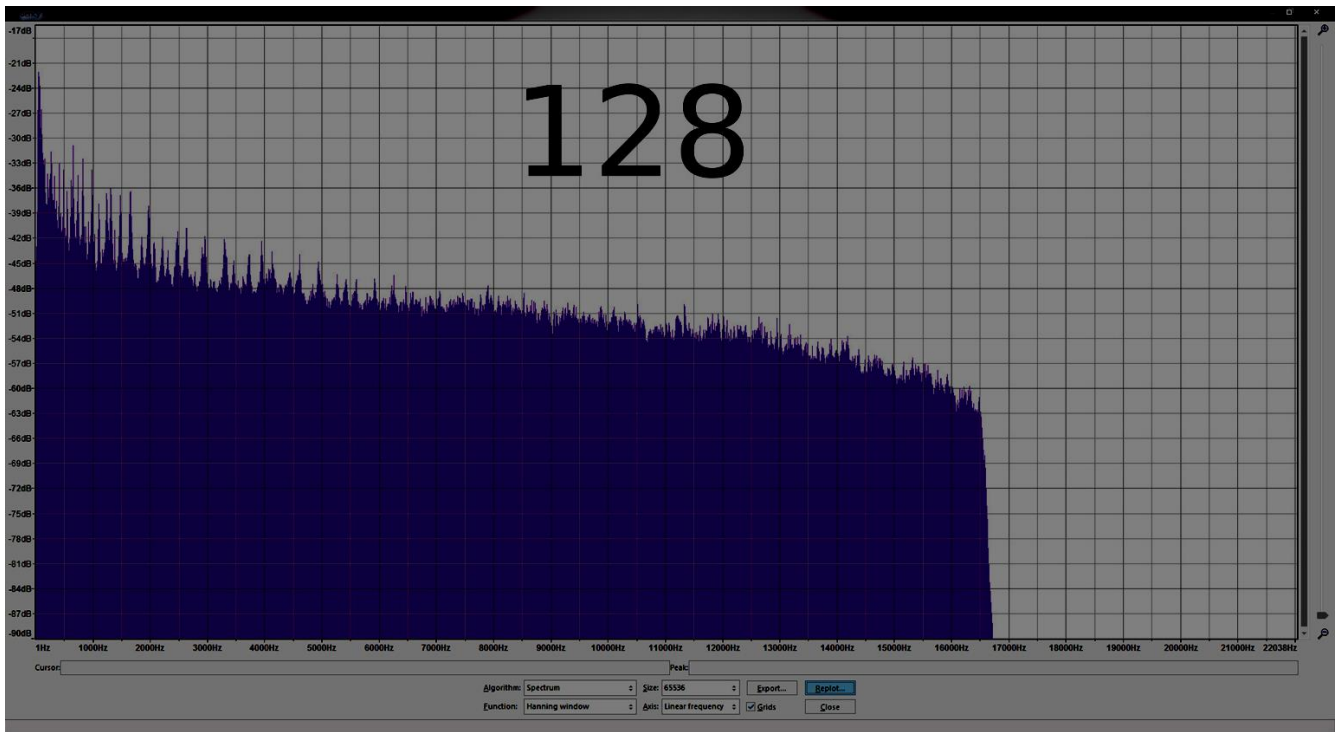
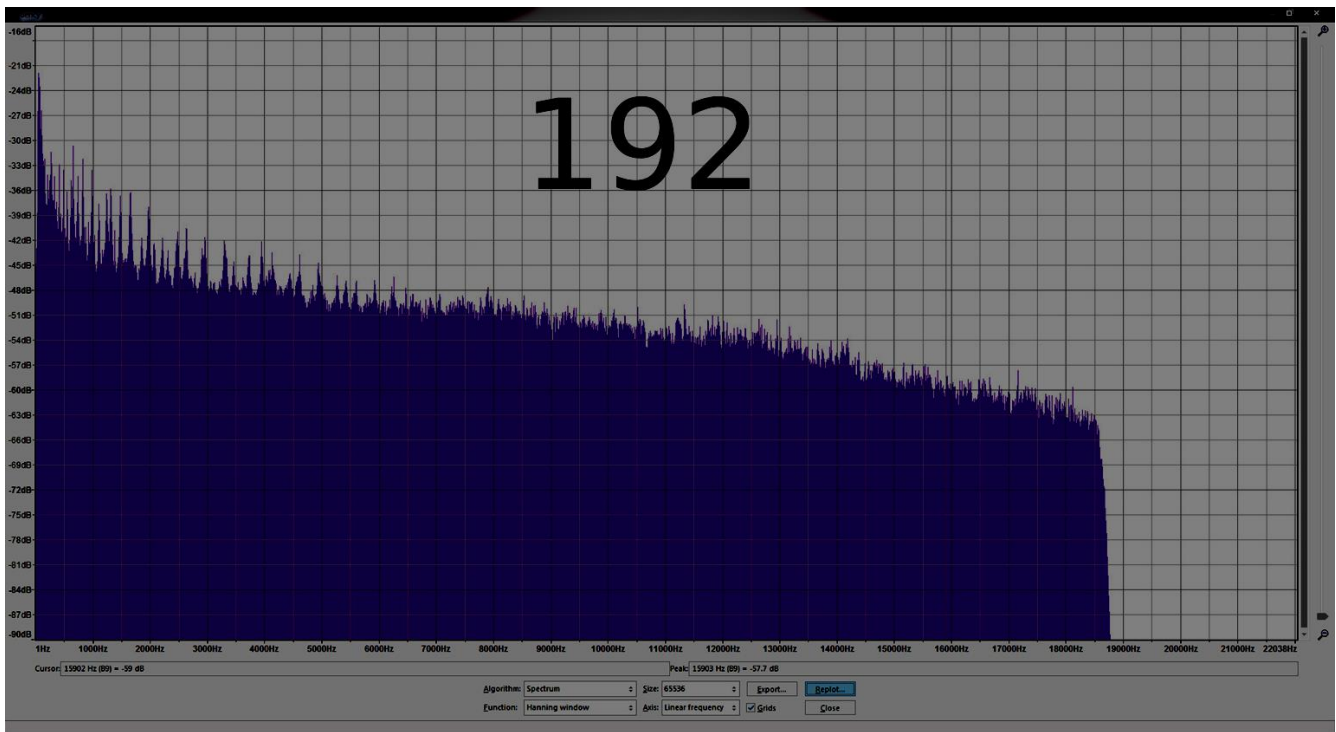
⁶ Assuming hearing that reaches 19000Hz

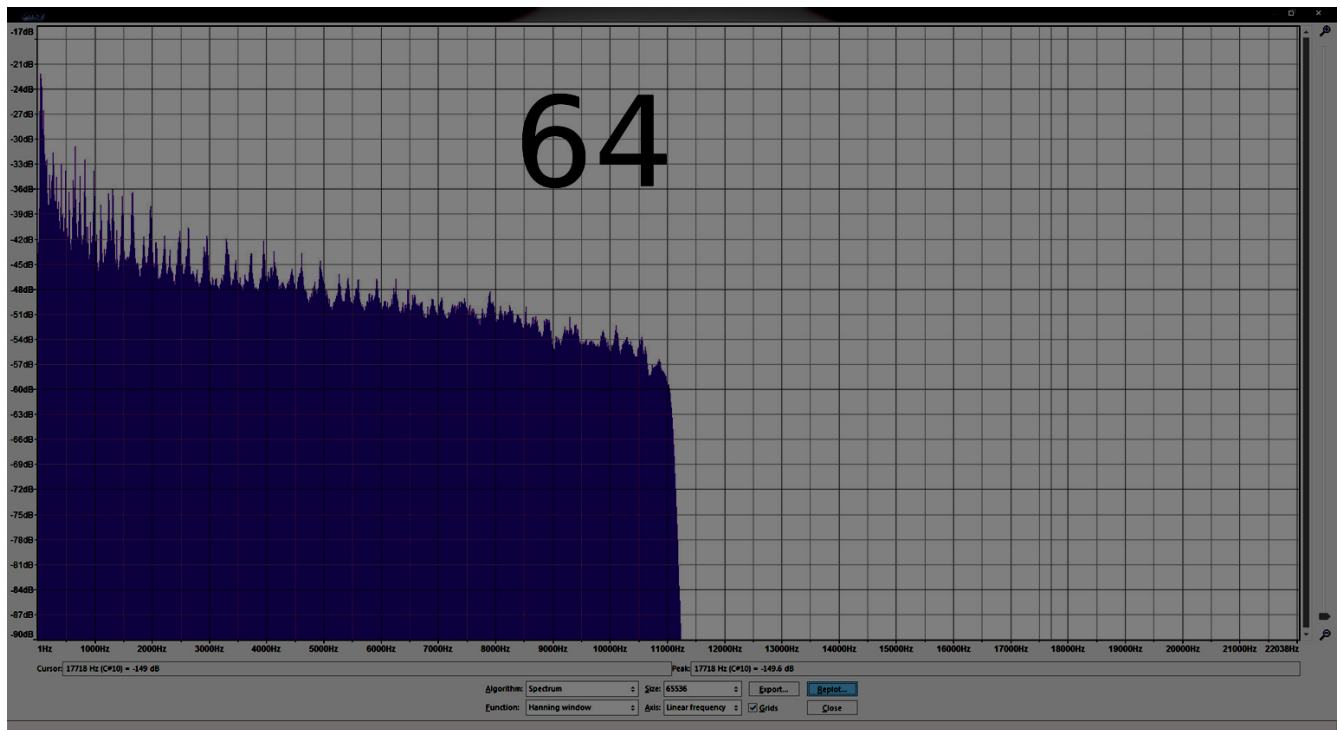
⁷ Can be perceived after prolonged hearing sessions under specific circumstances











VI. Analysis of Data Collected

The only visual difference between uncompressed and compressed music files was a cut-off at certain higher range frequencies in compressed music files. The same was true for all three music genres. For example, all content above 18.75 KHz is missing in 192 kbit/s MP3 files, while M4A files offer up to 19.75 KHz (more details on the exact frequencies can be seen in the previous section). In theory, these cut-offs alone should result in basically no distinguishable difference to the majority of human ears, especially when considering that most adults can not hear beyond 18 KHz. According to this theory, there would be no perceivable difference in sonic quality between a FLAC file and a 192 kbit/s MP3 file.

However, the subjective hearing test clearly revealed artifacts across all frequencies that are especially audible in the mid-range frequencies and become very pronounced in stronger compression

rates like 128 kbit/s MP3 files. It thus stands to reason that the MP3 compression algorithm has more effects than a cut-off in higher range frequencies. Assuming that this is not the case, one possible explanation can be found in harmonics. Removing higher range frequency content also eliminates higher harmonics. The loss of these higher harmonics could have a profound effect on the perceived sound signature and sonic quality of mid-range frequencies in compressed music. An identical tone can still be produced, even when removing certain harmonics, as long as the fundamental is still present. What does change though is the quality of that tone.

To test this theory, a low-pass filter was applied to the WAV version of 'Emoji', with the intention of imitating a 128 kbit/s MP3 file. The filter was applied at 16500Hz. During my hearing tests, the 128 kbit/s MP3 file instantly stood out with its poorer sonic quality compared to the uncompressed WAV file. The same should hold true for the artificially filtered WAV. However, the hearing test that followed revealed perfect clarity with no artifacts in the entire remaining frequency range. A frequency analysis of this modified file revealed a diagram that was in line with that of the reference WAV file and bore no resemblance to the diagram of the 128 kbit/s version of 'Emoji', with a full representation of all higher frequencies.

As such, the artifacts perceived in the mid-range frequencies must be the result of further compression techniques used in the complicated MP3 compression algorithm. This stretches beyond the scope of this project and was not pursued any further.

Despite what the numbers seemingly convey, the subjective hearing test bears slightly different results. As expected, there is no perceivable difference in sonic quality between WAV and FLAC files. In some real life cases, there can be a difference in perceived volume, but no such observation was made with the three test songs.

Because the theoretically perceivable difference is based on the presumption that only a cut-off of higher frequency content occurs in music file compression, there are some discrepancies between theory and observation. When comparing FLAC to MP3 320, no difference could be heard within a reasonable amount of time. That being said, differences can be perceived, not in higher range details (these details are beyond human hearing anyway in MP3 320), but instead in minuscule artifacts in mid-range frequencies that are a result of the MP3 compression algorithm. However, due to the relatively high bit rate, it is nearly impossible to hear these very subtle differences. It requires prolonged listening to music titles that are strongly familiar to the individual listening, as well as hardware capable of reproducing these nuances.

After this prolonged listening session, I notice fatigue in my ears that helps to reveal an ever so slight mushiness in mid-range frequencies (1-10KHz) in 320 kbit/s MP3 files that is not present in FLAC files. Due to the extremes that are necessary to perceive this difference, and the significantly higher amount of storage space required (22 vs. 77.25 GB, almost 4 times more space required), there is no reason for the average consumer to choose FLAC over MP3 320. That being said, it can depend on the type of MP3 compression (stereo vs. joint stereo, variable vs. constant) as well the amount of sonic content (more instruments), in which case the difference between MP3 320 and FLAC may become more distinct.

A real, much more relevant difference can be heard between 256 kbit/s MP3 and FLAC music. In this case, highs are very slightly recessed, leading to less details in 'hiss' sounds (percussion). Mids are mostly clear, with artifacts being barely noticable. Bass is still nuanced, accurate and tight-sounding. Considering the amount of space required, although less compared to MP3 320 (17.5 vs. 22

GB), it is not worth downgrading from MP3 320 to 256, as the space savings are nowhere near as extreme compared to FLAC vs. MP3 320.

64 and 128 kbit/s MP3 files can only be described, rather disgracefully, as sounding like ‘they were recorded underwater’. There is no detail in the highs, which are completely absent in MP3 64 to begin with, and mids are so full of artifacts that the aforementioned ‘underwater effect’ can be observed. Bass fares slightly better. It is present and discernible, but not accurate or punchy. 192 kbit/s is the first mainstream MP3 compression rate that offers decent sonic quality, with accurate bass and noticeable, but not disturbing, artifacts in mid range frequencies, with only the lack of clarity in higher frequencies truly letting it down.

A peculiar comparison is the M4A file and 256 kbit/s MP3 file. Both can be seen as direct competitors. The M4A was designed to offer better sonic quality. Indeed, bass is punchy and mids are surprisingly clear and warm sounding, being much more reminiscent of a 320 kbit/s MP3. However, although highs are extended by 250Hz more than on its MP3 competitor, the overall sound signature is still slightly more muddy than MP3 256. In terms of file size, M4A is bigger than MP3 320 (33.5 vs 22 GB), so it does not make sense to choose it over MP3 320 or possibly even MP3 256, which may explain why it was never able to replace MP3 files like it originally intended to.

VII. Discussion

The main difficulty of this project was the fact that there is no way to objectively measure something as subjective as sonic quality in music files. The only tangible measurable difference that can also be used as an objective bar of measurement for hearing is the high frequency content, or lack

thereof, in compressed music. As such, all subjective observations must be enjoyed with caution and are only valid for my own hearing. Other people may differ in critical listening skills and physical ability to pick up quality differences in music compression.

Furthermore, bass response was likely not measured very accurately. The LSR305 monitors will go down to 43Hz. At frequencies close to 43Hz, the LSR305 are not able to accurately reproduce an actual pitch, but rather tend to simply push air with no actual sonic quality. Although the DT880 Pro will reach 5Hz, human hearing is very insensitive to these low frequencies. Only a dedicated subwoofer could reveal potential differences in these low frequencies between different music compression rates, but such a device was not available for this project.

An area that could be further explored is MP3 compression itself. It was not the aim of this project, nor was it within its scope, but my tests did reveal that there are several, complicated, stages in the MP3 compression algorithm that could be disseminated and analyzed even further in a separate project.

Finally, the optimal file size vs. quality ratio appears to be the 320 kbit/s MP3. It shines with punchy and defined bass, no perceivable artifacts in mids in most real life scenarios and no perceivable difference in highs to most human ears. It emerges as the most sensible option to store music digitally.